THE UNIVERSITY OF ALABAMA

Day 1 - May 4th, 2026

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

B.Ed CS/Maths (Nigeria) | M.Sc CS (Fiji) |

MS, CS & PhD Candidate (United States)

Areas of specialization:

Empirical Software Engineering; Software Security

Software Engineering Education; Computer Science Education

Workshop Outline

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DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- Interactive Q & A
- IDE Setup and Resources

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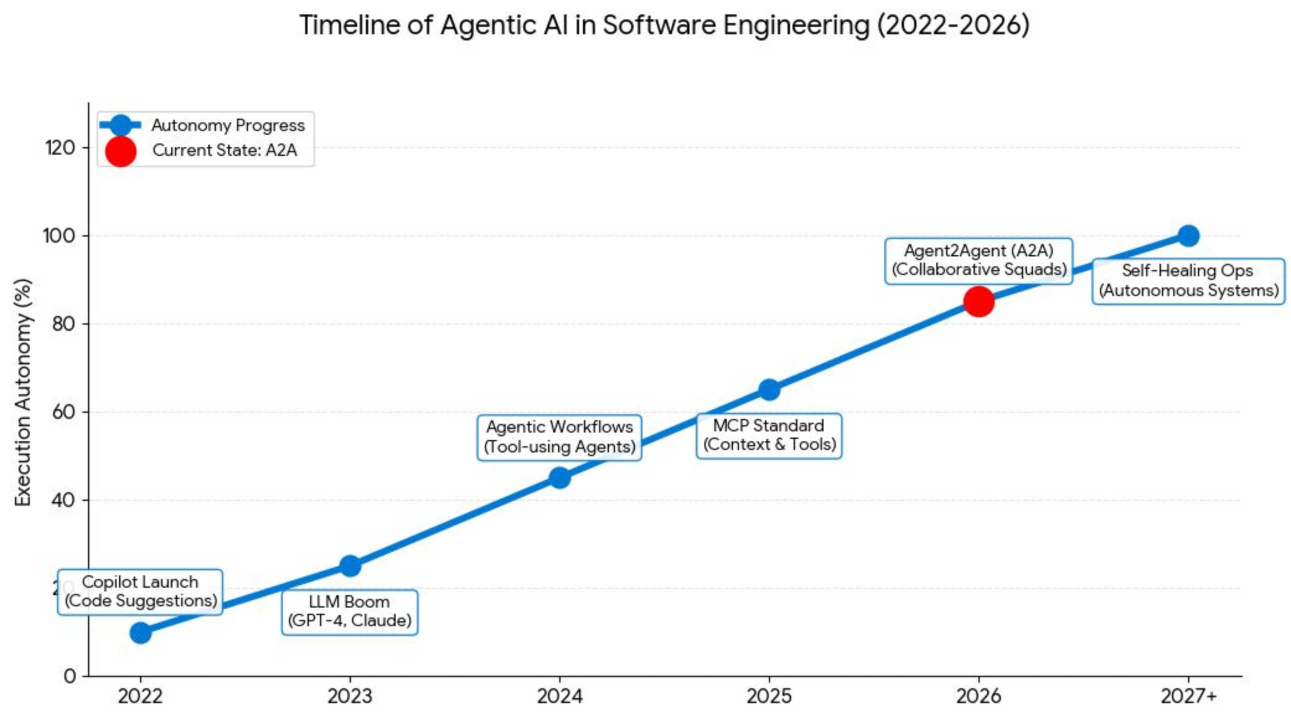
The Evolution of AI & Coding



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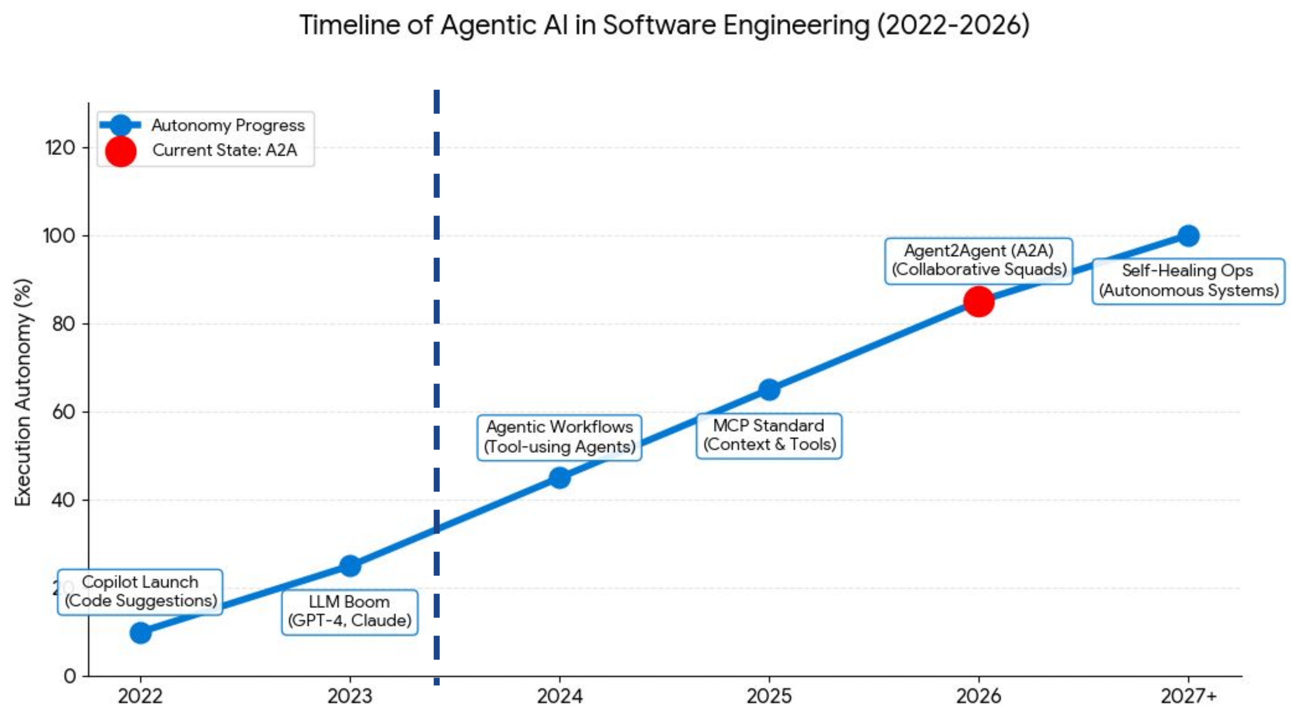

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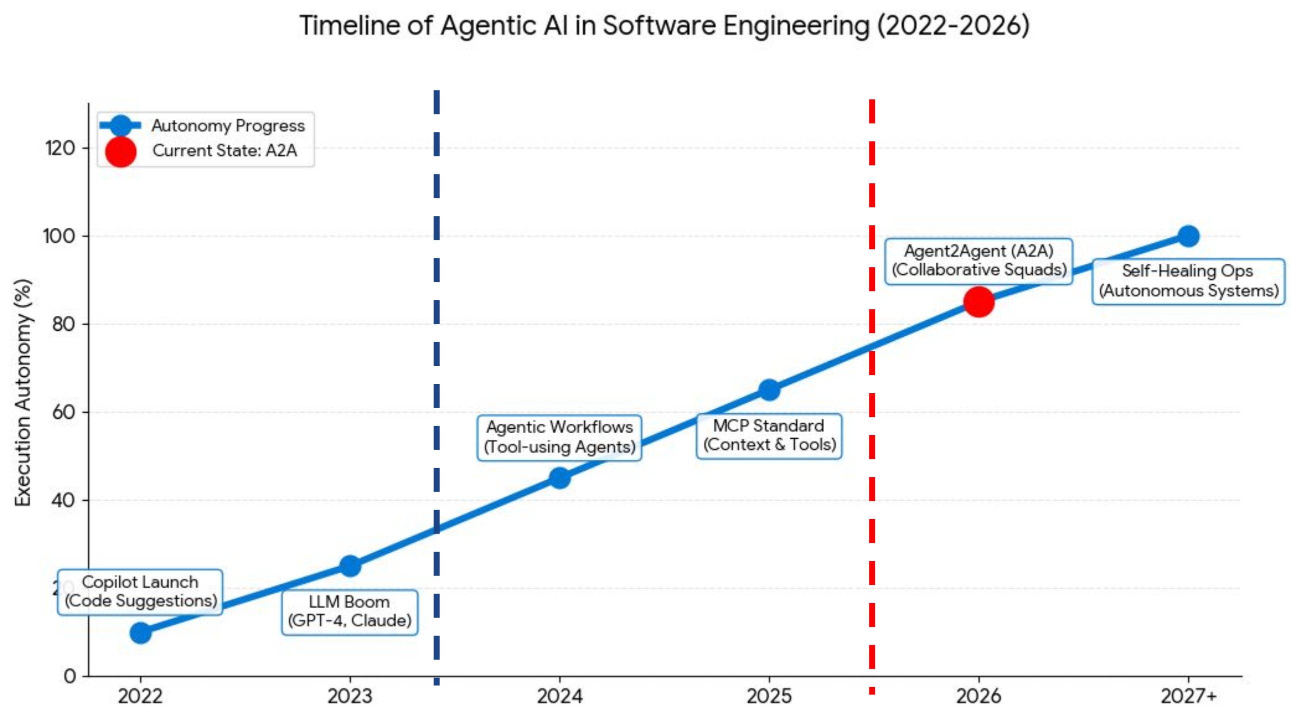

Why Mastering Agentic AI?

AI · CODING

An AI-powered coding tool wiped out a software company's database, then apologized for a 'catastrophic failure on my part'

BY **BEATRICE NOLAN**
TECH REPORTER

July 23, 2025 at 7:22 AM EDT

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Amjad Masad
@amasad

We saw Jason's post. @Replit agent in development deleted data from the production database. Unacceptable and should never be possible.

- Working around the weekend, we started rolling out automatic DB dev/prod separation to prevent this categorically. Staging environments in the works, too. More tomorrow.

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Why Mastering Agentic AI?

It took 9 seconds: AI agent running on Anthropic's Claude Opus 4.6 wipes critical database

Story by Business Today Desk • 19h • 3 min read

Source: https://www.msn.com/en-in/money/topstories/it-took-9-seconds-ai-agent-running-on-anthropic-s-claude-opus-4-6-wipes-critical-database/ar-AA21Np4C?ocid=socialshare

The Evolution of AI & Coding

1

Code Completion

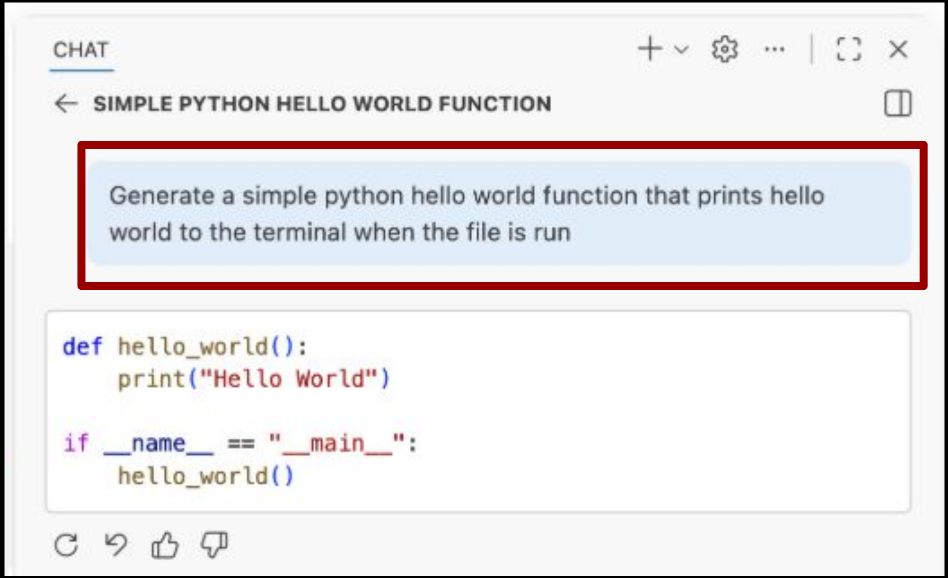
Intelligent snippets and autocomplete based on context. Tools like traditional IDE intellisense.

The Evolution of AI & Coding

2

Code Generation

Generative AI creating entire functions and boilerplate from natural language prompts. The era of Copilots.

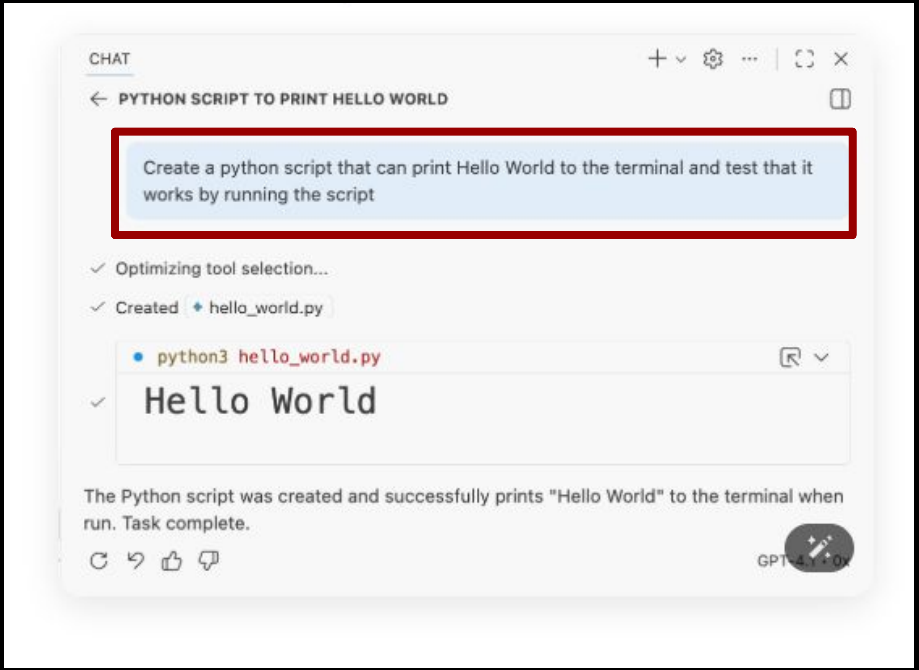


The Evolution of AI & Coding

3

Agentic Coding

Autonomous agents that reason, plan, and execute across files, using tools and verifying results.



Coding Agents & Demonstration

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

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CLI

- Copilot CLI
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- Pi

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Web

- Copilot in GitHub.com
- Codex
- Jules

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Custom

- Copilot SDK
- PyDantic AI
- LangChain

What is an AI Agent?

An AI Agent is an LLM that calls tools in a loop to achieve a goal.

Source: https://simonwillison.net/2025/Sep/18/agents/?utm_source=chatgpt.com

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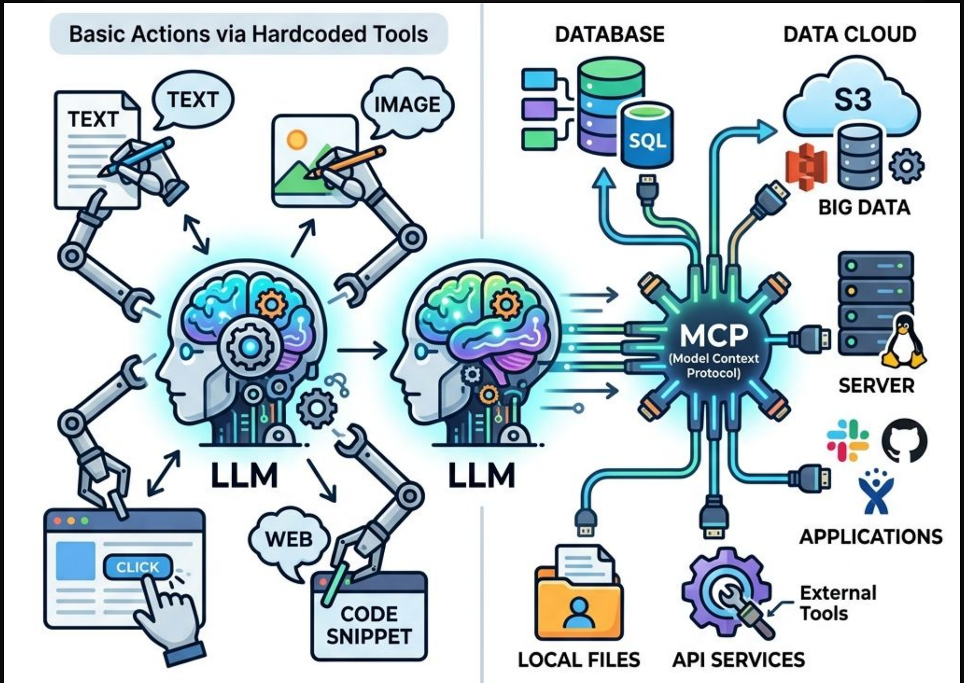

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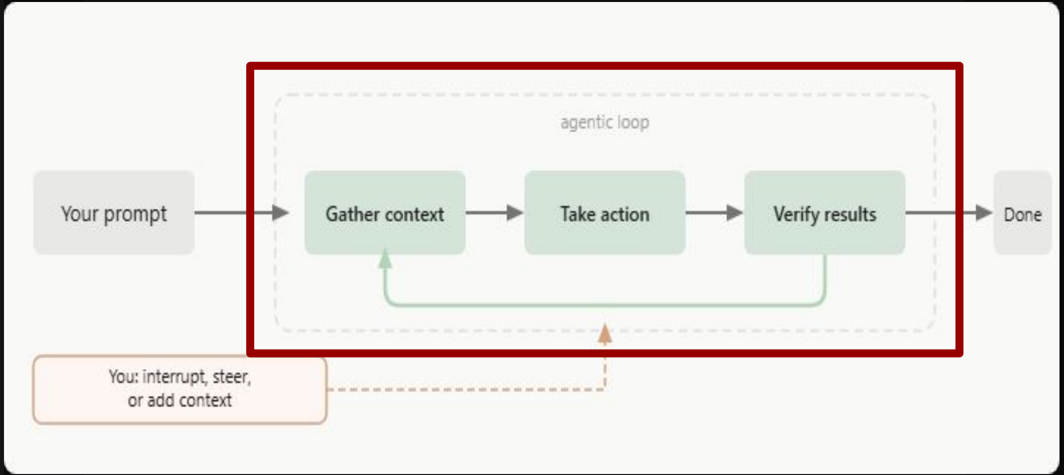

Source: https://code.claude.com/docs/en/how-claude-code-works

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Quick Pricing Overview (April 2026)

Service	Individual Plan	High-Usage/Max	Team/Business
GitHub Copilot	$10 /mo	$39 /mo (Pro+)	$19 /user/mo
Claude	$20 /mo	$100 - $200 /mo	$25 /user/mo
Antigravity	$29 /mo	$99 - $250 /mo	Custom
OpenAI Codex	$20 /mo (via ChatGPT Plus)	$200 /mo (Pro)	$25 /user/mo

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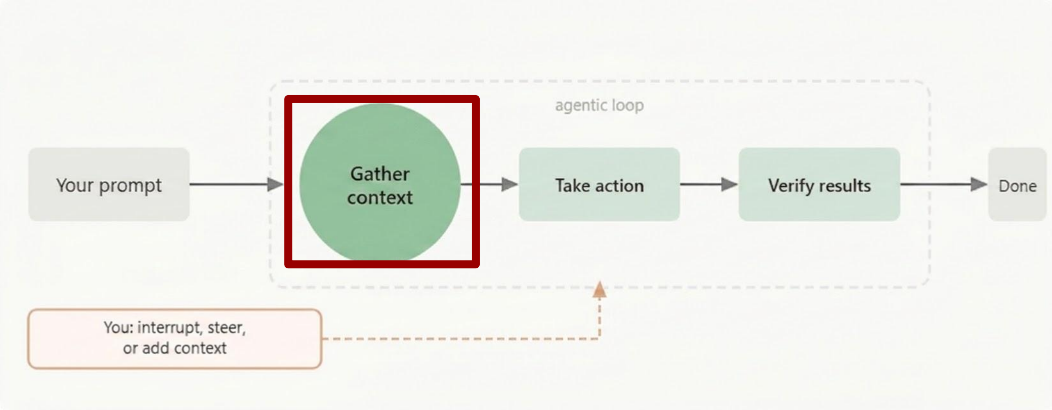
Equipping Agents with Context

Including Files and Assets

- Directly provide files, images, and GitHub items (Issues, PRs)
- Offer exemplar files containing best-practice code



Directive Documentation

- `copilot-instructions.md` — GitHub Copilot
- `cursor rules` — Specific to Cursor IDE
- `claude.md` — Tailored for Claude Code
- `agents.md` — The standard OpenAI format

Providing Tools to an Agent

MCP — Model Context Protocol

Open protocol for giving agents access to tools

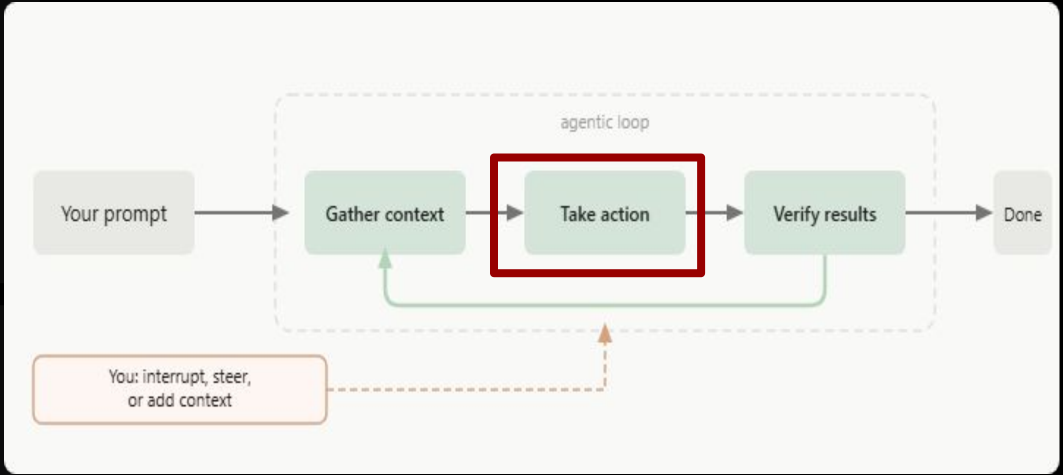
In IDE: search @mcp in extensions for vetted servers

Pro Tip: You can also create your own MCP servers

Agent Skills

A reusable workflow defined in a SKILL.MD file

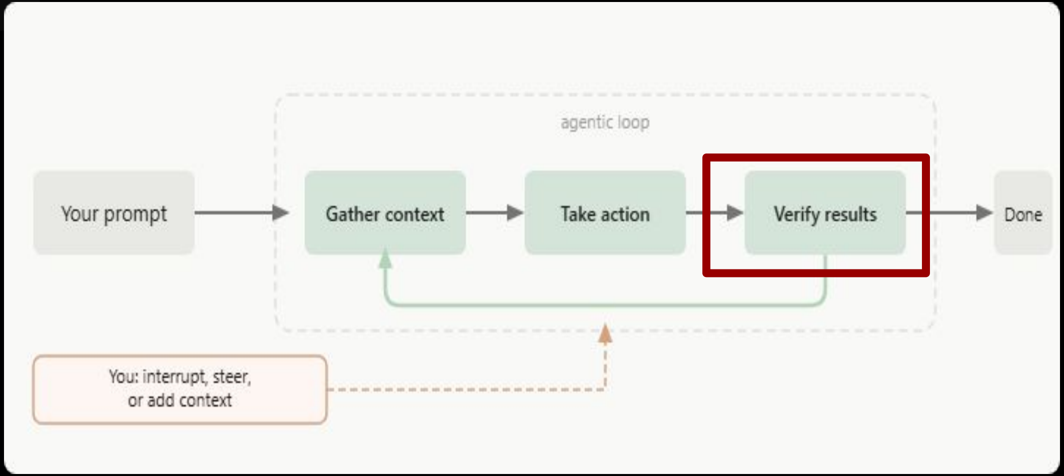
Skills can include tools, MCP servers, scripts, and more!



Verification Plan

Automated Tests - Playwright

Manual Verification



Resources

IDE (VS Code)

- VS Code Installation

  https://code.visualstudio.com/download

- Copilot Installation from VS Code extension

  https://www.youtube.com/watch?v=ws6t0NBle9g

- Launching Copilot for free (please you need to have a github account)

  https://www.youtube.com/watch?v=X_Aet9ndh_Y

IDE (Antigravity)

- Google Agent IDE Installation

  https://antigravity.google/download

You need to have gmail account to sign in and start using the agent for free

Resources

CLI (Copilot)

- Installation npm (requires Node.js 22+)

https://www.youtube.com/watch?v=BDxRhhs36ns

- Installation on windows via (WinGet)

https://www.youtube.com/watch?v=3mCaZ_tYdI0

CLI (Copilot)

Install commands:

Windows (WinGet): winget install GitHub.Copilot

npm (requires Node.js 22+): npm install -g @github/copilot

macOS / Linux: brew install copilot-cli

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Installation Support

Peer Support

Meet your colleagues for troubleshooting and technical guidance

OR

WhatsApp Support

Reach out to our dedicated support team on WhatsApp

Niyi: 0803-410-6706

Tossyno: 0703-448-6740

Prof. Dami: 0803-945-3555

DLaw: 0706-788-5378

Analyst: 0703-844-8548

Day 2 - May 5th, 2026

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The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

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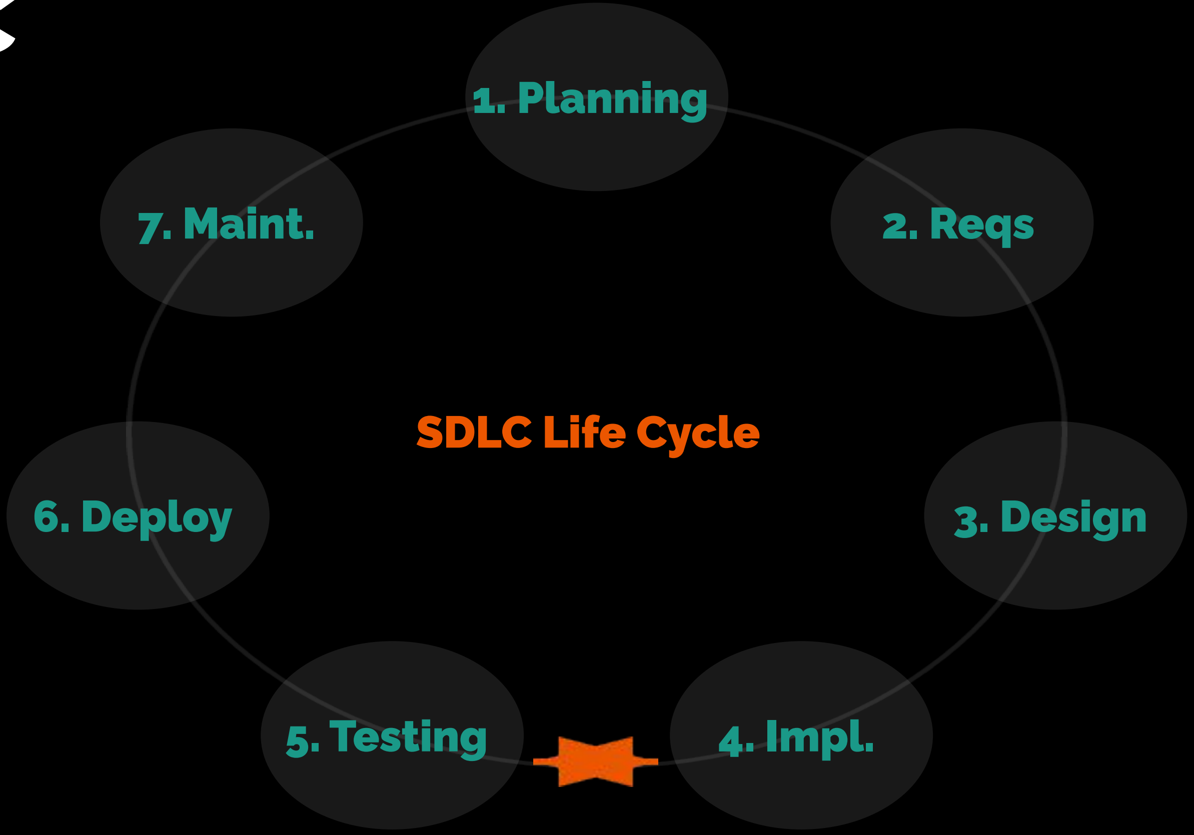
6. Deployment
Release to production environment.

SDLC


1. Planning
2. Reqs
3. Design
4. Impl.
5. Testing
6. Deploy
7. Maint.
SDLC Life Cycle

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Hands-on Learning

- For UX/UI

https://stitch.withgoogle.com/

Hands-on Learning

What app are you building? It’s time to build your desire app

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Define scope, costs, and identify risks.

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